STA2002 Probability and Statistics II
Assignment 6
Due date: **December 3rd 2023 11:59pm**
Please submit your written answers as a pdf file.

Please work on the assignment by yourself only. We follow a strict rule on academic honesty.

1. (ANOVA1) In this question, we will continue our investigation on the traffic volume dataset. Download the traffic dataset "traffic.csv" from Assignment 2 Blackboard folder. In particular, we would like to investigate the effect of weather conditions on the traffic volume. We shall only look at the effect of three weather: "Clouds", "Clear" and "Snow". These three weather conditions are the "treatments" in ANOVA. For this question, feel free to use any ANOVA function or package in your choice of programming language, and attach your code at the end of your assignment.

   (a) Fill in the following ANOVA table. If you decide to do it in Excel, it is really a few clicks away from the answer.

	SS	df	MS	F
Treatment				
Error				
Total				

   (b) Test the hypothesis that the hourly traffic has the same mean under these 3 weather conditions at $\alpha = 5\%$, assuming that the hourly traffic is normally distributed. State clearly (i) the degrees of freedom that you use in this F-test and (ii) the conclusion of the test.

(a)

	SS	df	MS	F
Treatment	2548511529	2	1274255764	338.3844
Error	1.18349E + 11	31428	3765705.1	
Total	1.20897E + 11	31430		

(b) The F statistic is $338.3844 > F_{0.05}(2, 31428) = 2.996$, so the null hypothesis that these 3 weather conditions have the same mean is rejected at 5%.

2. (ANOVA2) We would like to compare the lifetime of four brands of pens. It is suggested that the writing surface might affect lifetime, so the following lifetimes (in minutes) are collected for each brand-surface combination:

		Writing Surface		
		1	**2**	**3**
Brand of Pen	**1**	709,659	713,726	660,645
	2	668,685	722,740	692,720
	3	698,685	666,684	678,750
	4	698,650	704,666	686,733

(a) Fill in the following ANOVA table:

	SS	df	MS	F
Row				
Column				
Interaction				
Error				
Total				

(b) Is there an effect on the lifetime of pen due to the brand (i.e. row effect)? Conduct the appropriate test at $\alpha = 5\%$.

(c) Is there an effect on the lifetime of pen due to the writing surface (i.e. column effect)? Conduct the appropriate test at $\alpha = 5\%$.

(d) Is there an interaction effect on the lifetime of pen due to both the brand and the writing surface? Conduct the appropriate test at $\alpha = 5\%$.

(a)

	SS	df	MS	F
Row	1387.5	3	462.5	0.675511
Column	2888.083	2	1444.042	2.109116
Interaction	8100.25	6	1350.042	1.971823
Error	8216	12	684.6667	
Total	20591.83	23		

(b) From the ANOVA table, the F statistic is $0.675511 < 3.49 = F_{0.05}(3, 12)$, so the null hypothesis that there is no row effect is not rejected at 5%.

(c) From the ANOVA table, the F statistic is $2.109116 < 3.885 = F_{0.05}(2, 12)$, so the null hypothesis that there is no column effect is not rejected at 5%.

(d) From the ANOVA table, the F statistic is $1.971823 < 2.996 = F_{0.05}(6, 12)$, so the null hypothesis that there is no interaction effect is not rejected at 5%.

3. (Stock Index Changes 2011-2021) The data that you will use contains daily changes of six stock indices between 20112021. In this assignment, we will focus on the relationship between SP and UKX daily changes. We want to understand how the UKX daily changes (predictor) may explain the SP daily changes (response).

(a) Using a plot show the relationship between UKX and Sp. Comment on any pattern that you may see. Does a simple linear regression model seem reasonable?

(b) Fit and interpret the slope and intercept coefficients associated with the simple linear regression model.

(c) Discuss the meaning of the ANOVA F-test. Obtain the R^2 value and explain what this value means.

(d) Obtain 95% confidence intervals for the intercept and slope coefficients. Interpret your results.

(e) Under the 0.05 significance level, test whether the slope term is significantly different from zero. Write out the null and alternative hypotheses, the test statistic value, the p-value, the decision rule and your conclusion.

(f) Under the 0.05 significance level, test whether there is evidence to support that the slope term is significantly larger than 0.5. Write out the null and alternative hypotheses, the test statistic value, the p-value, the decision rule and your conclusion.

(g) Give the 95% Confidence and 95% Prediction Intervals when UKX is 0. Interpret these intervals.

(h) Look at relevant residual diagnostic plots and perform necessary test(s) to examine whether the model assumptions are satisfied. Please justify your conclusions.

(a) It is correct as long as you justify your conclusion by the pattern you observe.
Pattern1,The scatter plot of the data appears elliptical. In the case that the ellipse shape is very skinny and resembles a line, linear regression may be reasonable.
Pattern2, It concentrate to the origin, and the shape of the plot is more like a elliptic shape. If we apply linear regression to an elliptical dataset, the resulting line will not accurately represent the underlying relationship. The predicted values from the linear regression model will likely deviate significantly from the actual values, leading to poor model fit and inaccurate predictions.

(b) Slope: 0.64369 intercept:0.000457.
Interpretation: one unit of increase in UKX will increase SP by 0.64369. When UKX is 0 the SP will still have 0.000457.

(c) $R^2 = 0.3749$. This means that the regression model can explain about 37.49% of the variance.

(d) Intercept: [5.808e-05,8.567e-04] coefficient:[6.050e-01,6.824e-01].
Interpretation of the confidence interval for the intercept: The confidence interval [5.808e-05, 8.567e-04] suggests that, based on the data and the model used, we can be confident that the true population intercept falls within this range. This means that if we were to repeat the study or collect new data, the estimated intercept would likely fall between 5.808e-05 and 8.567e-04 in approximately 95% of the cases.
Interpretation of the confidence interval for the coefficient: The confidence interval [6.050e-01, 6.824e-01] indicates that, with a certain level of confidence, the true population coefficient lies within this range. It suggests that the estimated coefficient for the given independent variable falls between 6.050e-01 and 6.824e-01 in approximately 95% of the cases if we were to repeat the study or collect new data.

(e) $H_0 : \beta_1 = 0$, $H_1 : \beta_1 \neq 0$ $T == 32.637$ p-value$\simeq 0$. We reject H_0 by the decision rule that p¡0.05.

(f) $H_0 : \beta_1 = 0.5$, $H_1 : \beta_1 > 0.5$ $T = 7.2857$ p-value$\simeq 0$. We reject H_0 by the decision rule that $p < 0.05$.

(g) $[5.808e-05, 8.567e-04]$ The confidence interval [5.808e-05, 8.567e-04] suggests that, based on the data and the model used, we can be confident that the true population intercept falls within this range. This means that if we were to repeat the study or collect new data, the estimated intercept would likely fall between 5.808e-05 and 8.567e-04 in approximately 95% of the cases.

[-0.016378, 0.017293] if we were to make a prediction using the linear regression model, the true value of the response variable for a new observation would be expected to fall between -0.016378 and 0.017293 with 95% of confidence.

(h) By the durbinWatson test, the assumption is not satisfied.

4. (Properties of k_i) show that for

$$k_i = \frac{X_i - \bar{X}}{\sum_{i=1}^{n} \left(X_i - \bar{X}\right)^2}$$

the following holds

(a)

$$k_1 + \ldots + k_n = 0$$

(b)

$$k_1 X_1 + \ldots + k_n X_n = 1$$

(c)

$$k_1^2 + \ldots + k_n^2 = \frac{1}{\sum_{i=1}^{n} \left(X_i - \bar{X}\right)^2}$$

(a) $\sum_{i=1}^{n} k_i = \sum_{i=1}^{n} \frac{X_i - \bar{X}}{\sum_{i=1}^{n} \left(X_i - \bar{X}\right)^2} = \frac{\sum_{i=1}^{n} X_i - \bar{X}}{\sum_{i=1}^{n} \left(X_i - \bar{X}\right)^2} == \frac{n\bar{X} - n\bar{X}}{\sum_{i=1}^{n} \left(X_i - \bar{X}\right)^2} = 0$

(b) $\sum_{i=1}^{n} k_i X_i = \sum_{i=1}^{n} \frac{X_i^2 - X_i \bar{X}}{\sum_{i=1}^{n} \left(X_i - \bar{X}\right)^2} = \frac{\sum_{i=1}^{n} X_i^2 - n\bar{X}^2}{\sum_{i=1}^{n} \left(X_i - \bar{X}\right)^2} = 1$

(c) $\sum_{i=1}^{n} k_i^2 = \sum_{i=1}^{n} \left(\frac{X_i - \bar{X}}{\sum_{i=1}^{n} \left(X_i - \bar{X}\right)^2} \right)^2 = \sum_{i=1}^{n} \frac{(X_i - \bar{X})^2}{(\sum_{i=1}^{n} \left(X_i - \bar{X}\right)^2)^2} = \frac{\sum_{i=1}^{n} (X_i - \bar{X})^2}{(\sum_{i=1}^{n} \left(X_i - \bar{X}\right)^2)^2} = \frac{1}{\sum_{i=1}^{n} \left(X_i - \bar{X}\right)^2}$